

Code ▼

Keio - Esophageal Neo-FLOT_Clinical Final analysis 05202026

```
library(swimplot) library(coxphf) library(grid) library(gtable) library(readr) library(mosaic) library(dplyr) library(survival)
library(survminer) library(ggplot2) library(scales) library(coxphf) library(ggthemes) library(tidyverse)
library(gtsummary) library(flextable) library(parameters) library(car) library(ComplexHeatmap) library(tidyverse)
library(readxl) library(survival) library(janitor) library(openxlsx) library(writexl) library(rms) library(pROC) library(DT)
```

#ctDNA Detection rate by Stage and Window

Hide

```
#Baseline
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.Base <- factor(circ_data$ctDNA.Base, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III","IV"))
circ_data <- subset(circ_data, ctDNA.Base %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.Base == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.Base, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.Base == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

	Stage	Total_Count	Positive_Count	Rate
1	I	3	2	66.67%
2	II	7	7	100.00%
3	III	19	19	100.00%
4	IV	5	5	100.00%
5	Overall	34	33	97.06%

Hide

```
#post-NAC Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.Base <- factor(circ_data$ctDNA.postNAC, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III","IV"))
circ_data <- subset(circ_data, ctDNA.postNAC %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.postNAC == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.postNAC, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.postNAC == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

	Stage	Total_Count	Positive_Count	Rate
1	I	3	1	33.33%
2	II	7	2	28.57%
3	III	18	7	38.89%
4	IV	5	4	80.00%
5	Overall	33	14	42.42%

Hide

```

#MRD Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III"))
circ_data <- subset(circ_data, ctDNA.MRD %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.MRD == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.MRD, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.MRD == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)

```

	Stage	Total_Count	Positive_Count	Rate
1	I	3	0	0.00%
2	II	6	1	16.67%
3	III	16	7	43.75%
4	Overall	29	9	31.03%

#Demographics Table

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data_subset <- circ_data %>%
  select(
    Age,
    Sex,
    ECOG,
    Location,
    Stage,
    Resection,
    pGrade,
    Recurrence,
    OS.Event,
    OS.months) %>%
  mutate(
    Age = as.numeric(Age),
    Sex = factor(Sex, levels = c("Male", "Female")),
    ECOG = factor(ECOG, levels = c("0", "1")),
    Location = factor(Location, levels = c("Lt", "Mt", "Ut")),
    Stage = factor(Stage, levels = c("I","II","III","IV")),
    Resection = factor(Resection, levels = c("R0","R1")),
    pGrade = factor(pGrade, levels = c("Grade0","Grade1","Grade2","Grade3")),
    Recurrence = factor(Recurrence, levels = c("TRUE","FALSE")),
    OS.Event = factor(OS.Event, levels = c("TRUE","FALSE")),
    OS.months = as.numeric(OS.months))
table1 <- circ_data_subset %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{median} ({min} - {max})",
      all_categorical() ~ "{n} ({p}%)" ) %>%
    bold_labels()
  )
table1
```

Characteristic	N = 34 ¹
Age	66 (44 - 80)
Sex	
Male	27 (79%)
Female	7 (21%)
ECOG	
0	32 (94%)
1	2 (5.9%)
¹ Median (Min - Max); n (%)	

Characteristic	N = 34¹
Location	
Lt	14 (41%)
Mt	18 (53%)
Ut	2 (5.9%)
Stage	
I	3 (8.8%)
II	7 (21%)
III	19 (56%)
IV	5 (15%)
Resection	
R0	29 (94%)
R1	2 (6.5%)
Unknown	3
pGrade	
Grade0	1 (3.2%)
Grade1	15 (48%)
Grade2	12 (39%)
Grade3	3 (9.7%)
Unknown	3
Recurrence	
TRUE	18 (53%)
FALSE	16 (47%)
OS.Event	
TRUE	12 (35%)
FALSE	22 (65%)
OS.months	26 (8 - 52)
¹ Median (Min - Max); n (%)	

Hide

```
fit1 <- as_flex_table(  
  table1,  
  include = everything(),  
  return_calls = FALSE,  
  strip_md_bold = TRUE)  
fit1
```

Characteristic	N = 34 ¹
Age	66 (44 - 80)
Sex	
Male	27 (79%)
Female	7 (21%)
ECOG	
0	32 (94%)
1	2 (5.9%)
Location	
Lt	14 (41%)
Mt	18 (53%)
Ut	2 (5.9%)
Stage	
I	3 (8.8%)
II	7 (21%)
III	19 (56%)
IV	5 (15%)
Resection	
R0	29 (94%)
R1	2 (6.5%)
Unknown	3
pGrade	
Grade0	1 (3.2%)
Grade1	15 (48%)
Grade2	12 (39%)
Grade3	3 (9.7%)

¹Median (Min - Max); n (%)

Characteristic	N = 34 ¹
Unknown	3
Recurrence	
TRUE	18 (53%)
FALSE	16 (47%)
OS.Event	
TRUE	12 (35%)
FALSE	22 (65%)
OS.months	26 (8 - 52)

¹Median (Min - Max); n (%)

Hide

```
save_as_docx(fit1, path= "~/Downloads/1. Keio ESCC_Demographics Table.docx")
```

#Heatmap with Clinical & Genomics Factors

Hide

```

rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data %>% arrange(Stage)
circ_datadf <- as.data.frame(circ_data)

ha <- HeatmapAnnotation(
  Stage = circ_data$Stage,
  Sex = circ_data$Sex,
  Location = circ_data$Location,
  Grade = circ_data$Grade,
  ctDNA.Base = circ_data$ctDNA.Base,
  ctDNA.postNAC = circ_data$ctDNA.postNAC,
  ctDNA.MRD = circ_data$ctDNA.MRD,
  ctDNA.surveillance = circ_data$ctDNA.surveillance,
  EFS.Event = circ_data$EFS.Event,
  OS.Event = circ_data$OS.Event,

  col = list(Stage = c("I" = "seagreen1", "II" = "seagreen", "III" = "orange", "IV" = "purple"),
    Sex = c("Female" = "goldenrod", "Male" = "blue4"),
    Location = c("Lt" = "brown", "Mt" = "darkgreen", "Ut" = "orange4"),
    Grade = c("Grade1b/2/3" = "cornflowerblue", "Grade0/1a" = "darkmagenta"),
    ctDNA.Base = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.postNAC = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.MRD = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    EFS.Event = c("TRUE" = "red3", "FALSE" = "blue"),
    OS.Event = c("TRUE" = "black", "FALSE" = "gray")
  ))
ht <- Heatmap(matrix(nrow = 0, ncol = length(circ_data$Stage)), show_row_names = FALSE, cluster_rows = F, cluster_columns = FALSE, top_annotation = ha)
pdf("heatmap.pdf", width = 7, height = 7)
draw(ht, annotation_legend_side = "bottom")
dev.off()

```

```

null device
      1

```

#Overview Plot

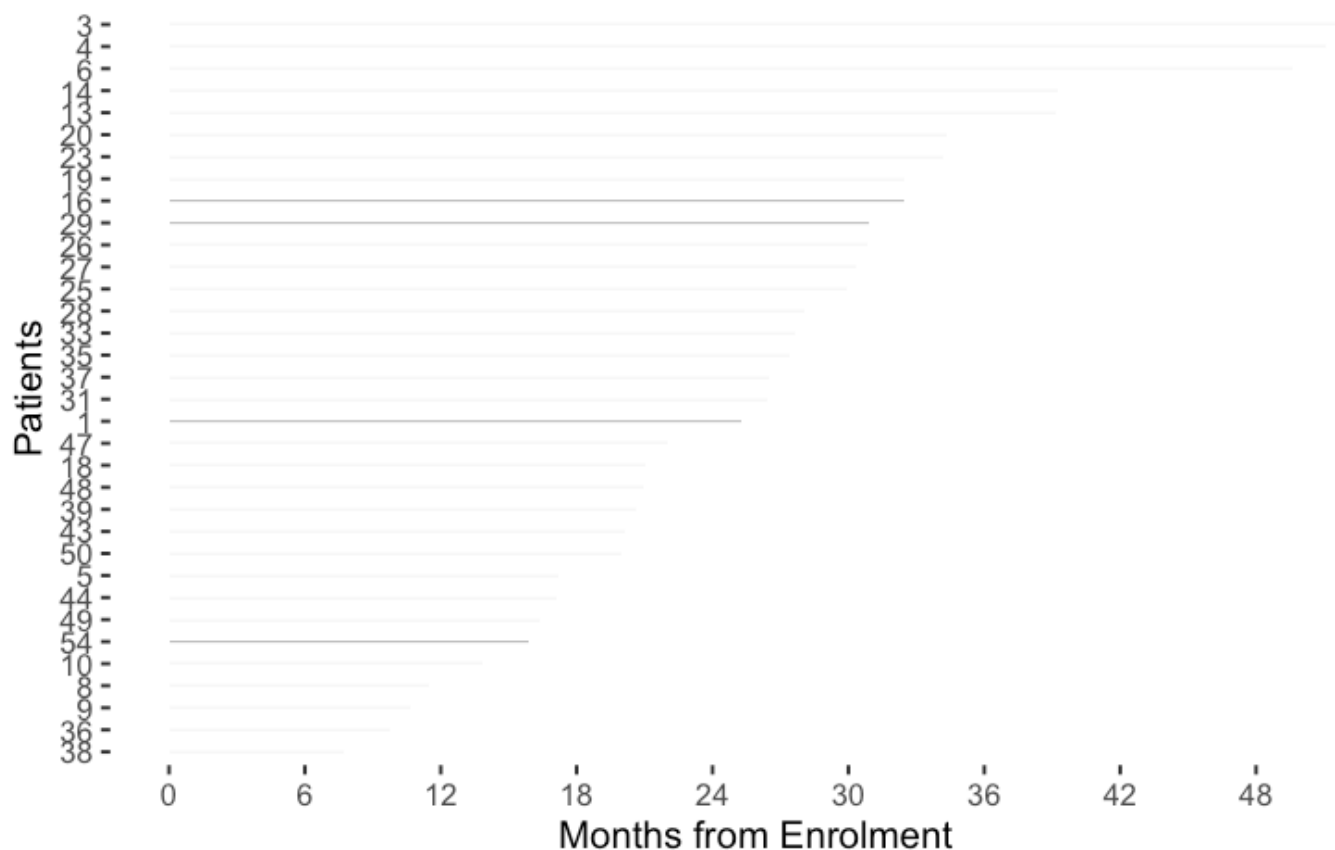
Hide


```

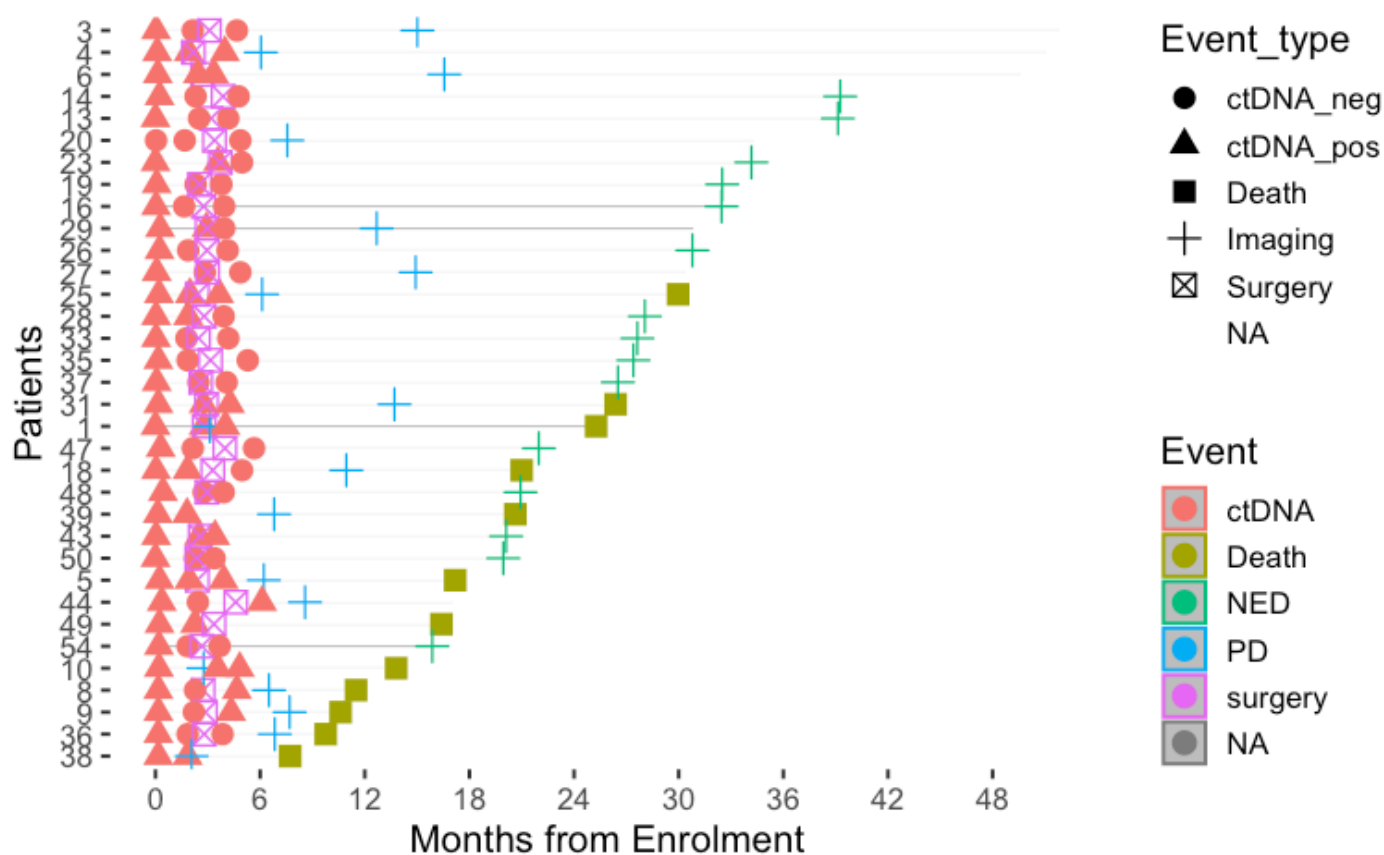
setwd("~/Downloads")
clinstage<- read.csv("Keio Neo FLOT_OP.csv")
clinstage_df<- as.data.frame(clinstage)

#Display the swimmer plot with the label box
oplot<-swimmer_plot(df=clinstage_df,
                    id='PatientName',
                    end='fu.diff.months',
                    fill='gray',
                    width=.01,
                    base_size = 14)
oplot <- oplot + theme(panel.border = element_blank())
oplot <- oplot + scale_y_continuous(breaks = seq(-12, 48, by = 6))
oplot <- oplot + labs(x ="Patients" , y="Months from Enrolment")
oplot

```


[Hide](#)

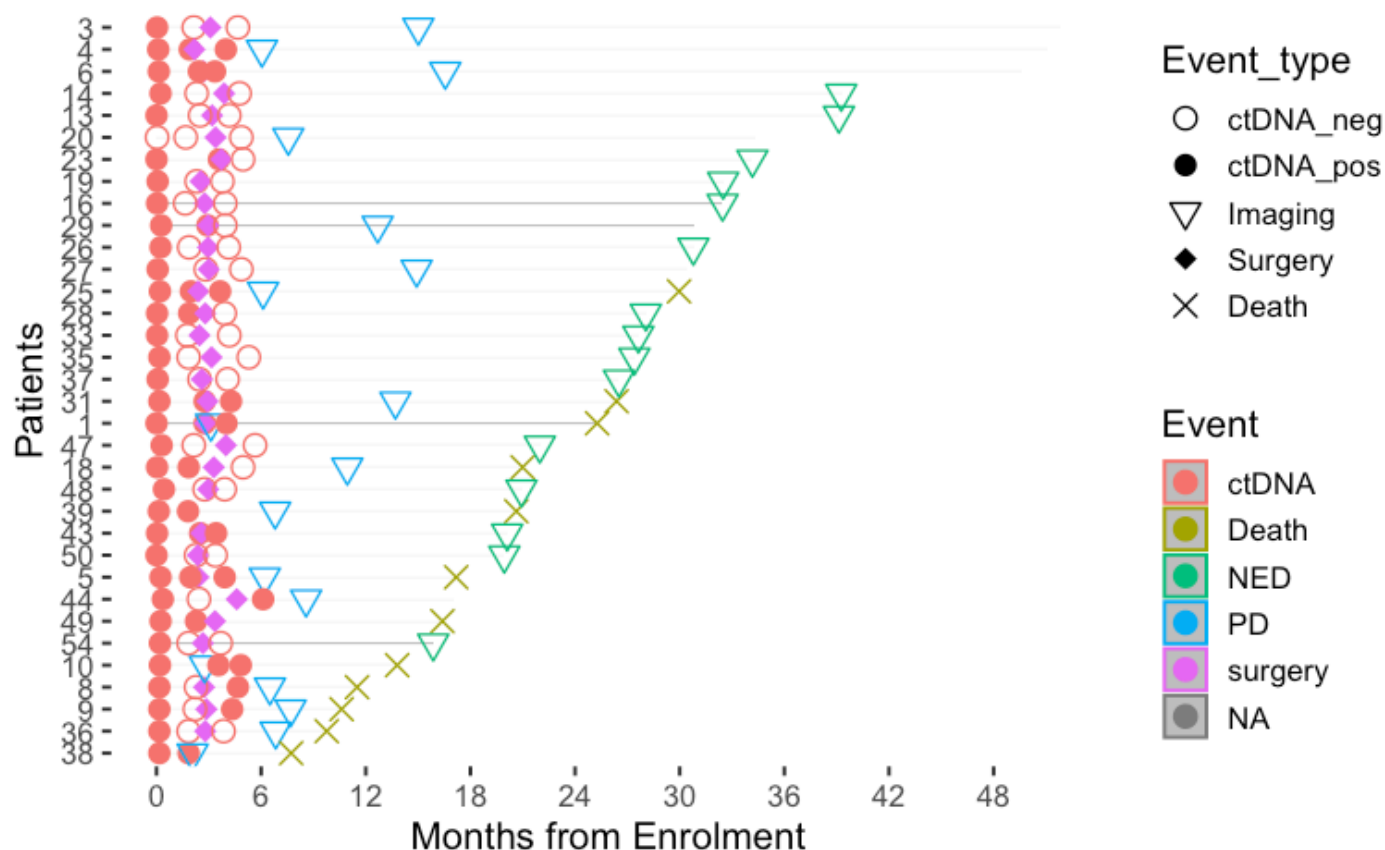
```
##plot events
oplot_ev1 <- oplot + swimmer_points(df_points=clinstage_df,
                                   id='PatientName',
                                   time='date.diff.months',
                                   name_shape = 'Event_type',
                                   name_col = 'Event',
                                   size=3.5,fill='black',
                                   #col='darkgreen'
)
oplot_ev1
```


[Hide](#)

```
#Shape customization to Event_type

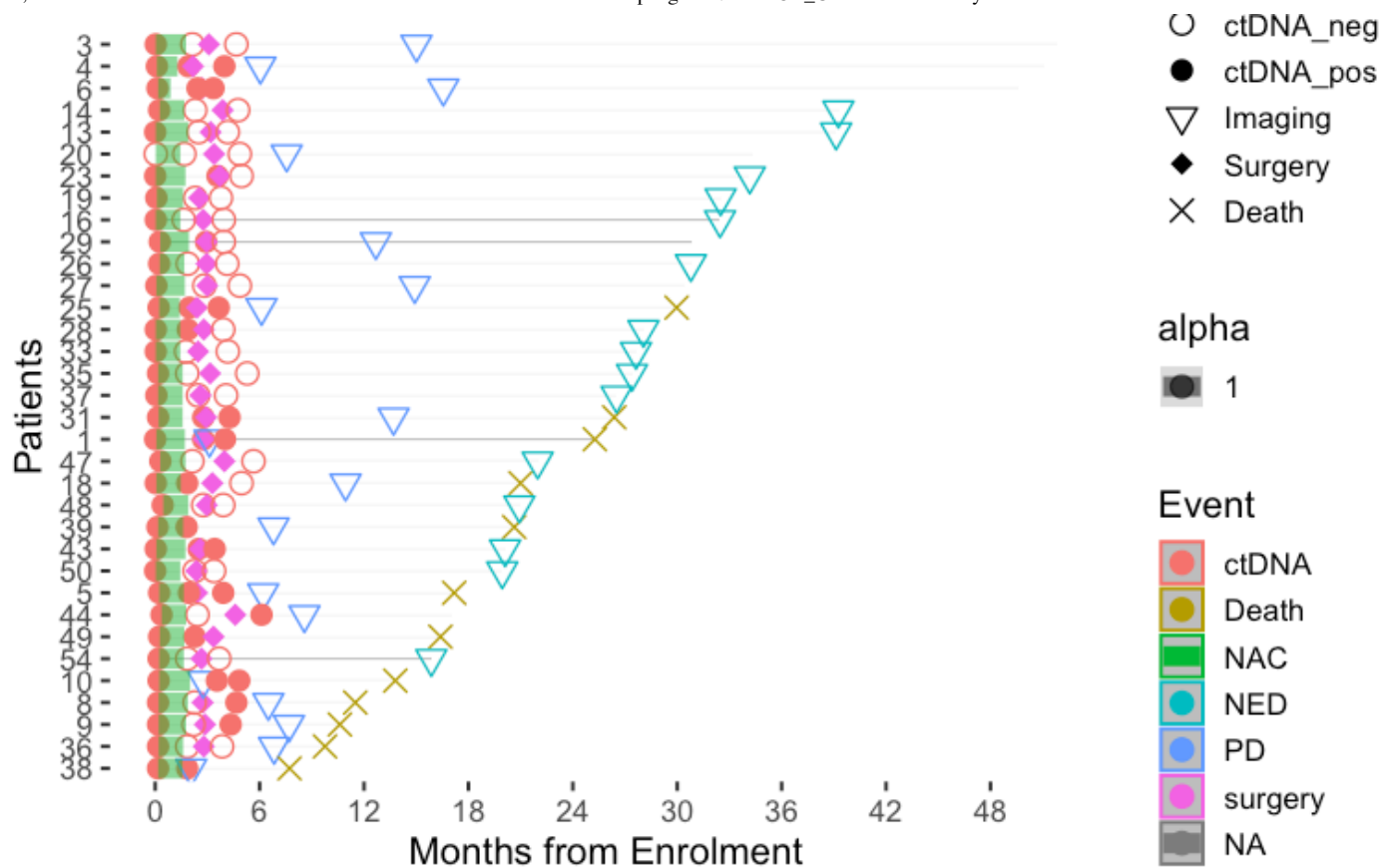
oplot_ev1.1 <- oplot_ev1 + ggplot2::scale_shape_manual(name="Event_type",values=c(1,16,
6,18,4),breaks=c('ctDNA_neg','ctDNA_pos', 'Imaging','Surgery', 'Death'))

oplot_ev1.1
```

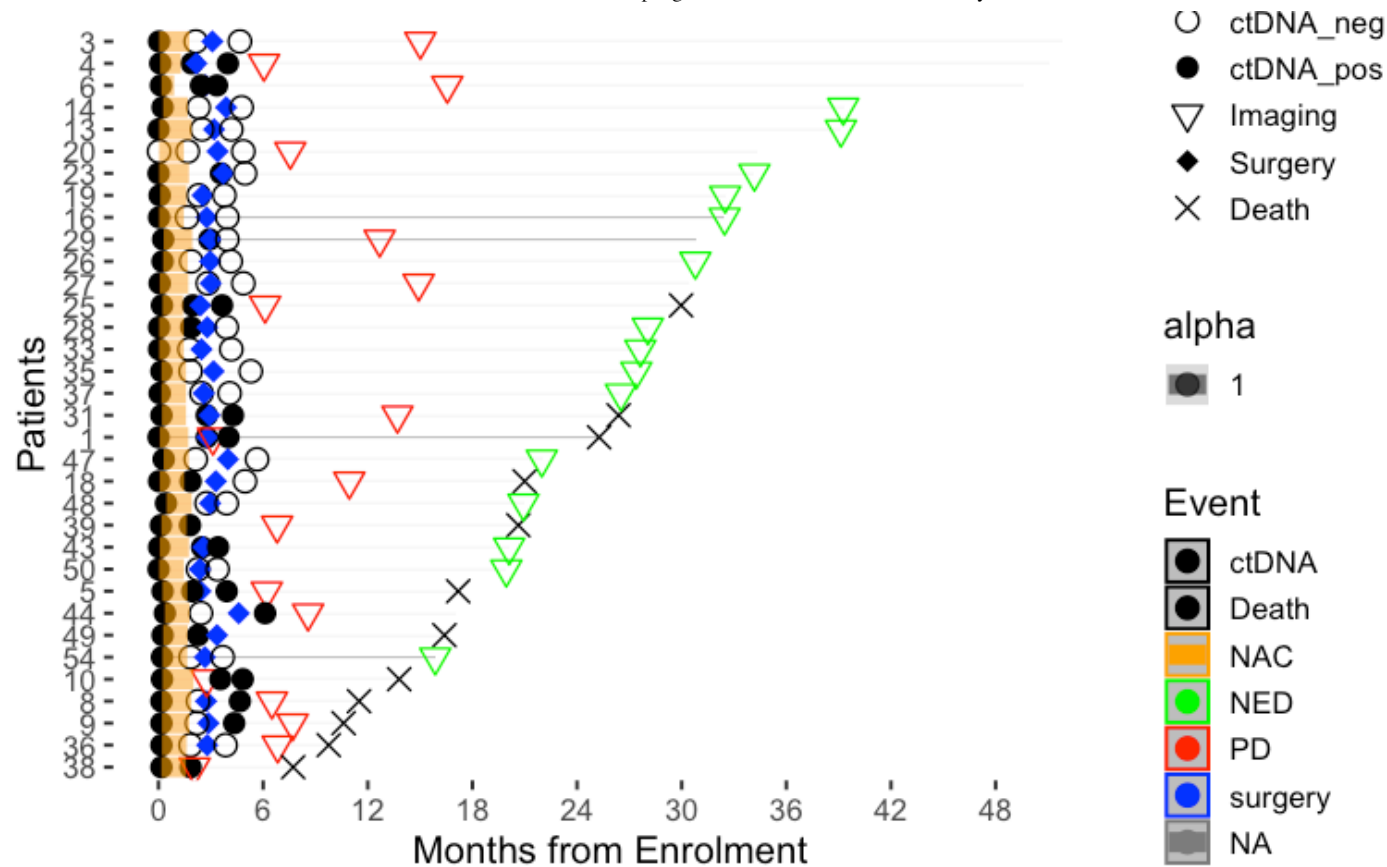

[Hide](#)

```
#plot treatment
```

```
oplot_ev2 <- oplot_ev1.1 + swimmer_lines(df_lines=clinstage_df,
                                         id='PatientName',
                                         start='Tx_start.months',
                                         end='Tx_end.months',
                                         name_col='Tx_type',
                                         size=3.5,
                                         name_alpha = 1.0)
oplot_ev2 <- oplot_ev2 + guides(linetype = guide_legend(override.aes = list(size = 5, color = "black")))
oplot_ev2
```


[Hide](#)

```
#colour customization
oplot_ev2.2 <- oplot_ev2 + ggplot2::scale_color_manual(name="Event", values=c("black", "black", "orange", "green", "red", "blue"))
oplot_ev2.2
```



#EFS by ctDNA post-NAT - all stages

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.postNAC!="",]
circ_datadf <- as.data.frame(circ_data)

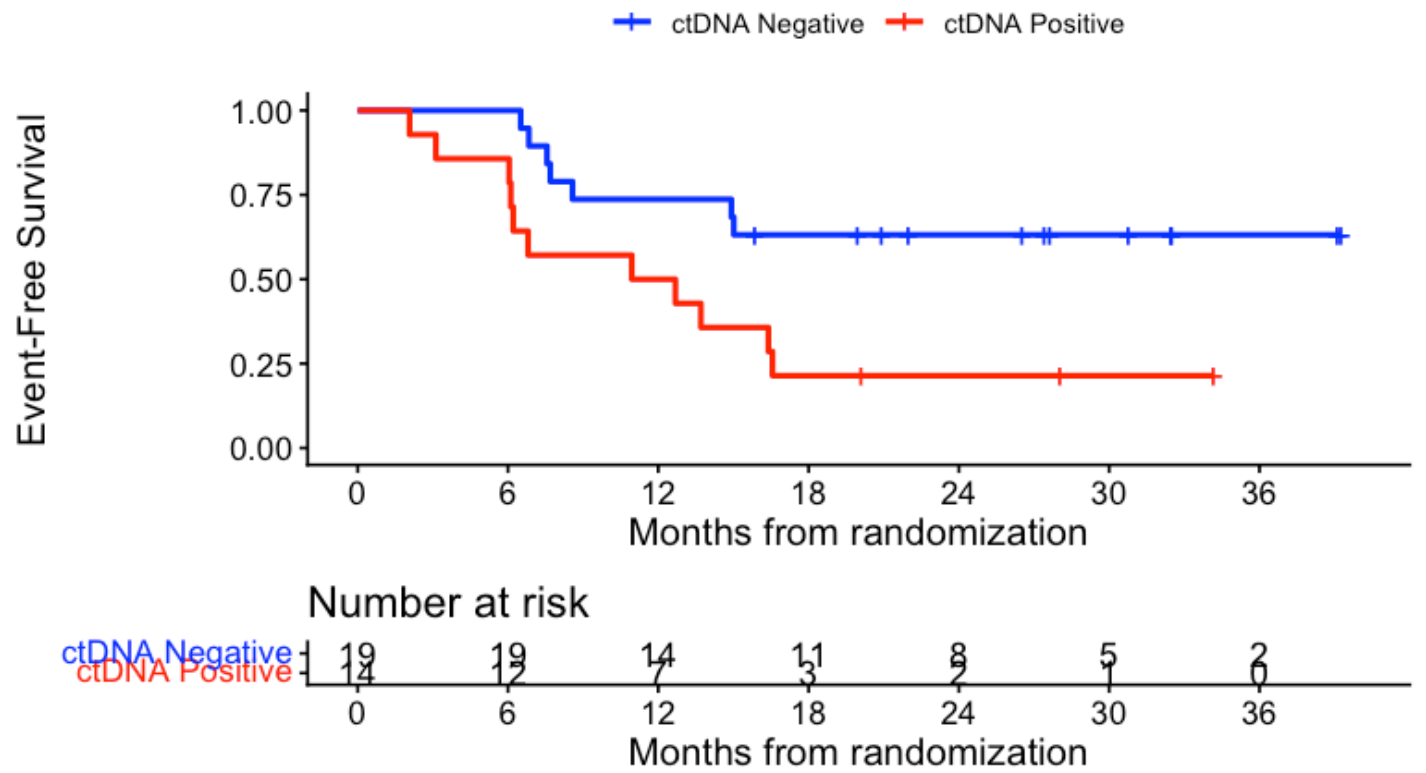
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.postNAC, data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$EFS.months, event = circ_data\$EFS.Event) ~ ctDNA.postNAC, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.postNAC=NEGATIVE	19	7	NA	15.01	NA
ctDNA.postNAC=POSITIVE	14	11	11.8	6.21	NA

```
surv_object <-Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.postNAC, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA status post-NAT | All pts", ylab= "Event-Free Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```

EFS - ctDNA status post-NAT | All pts



Hide

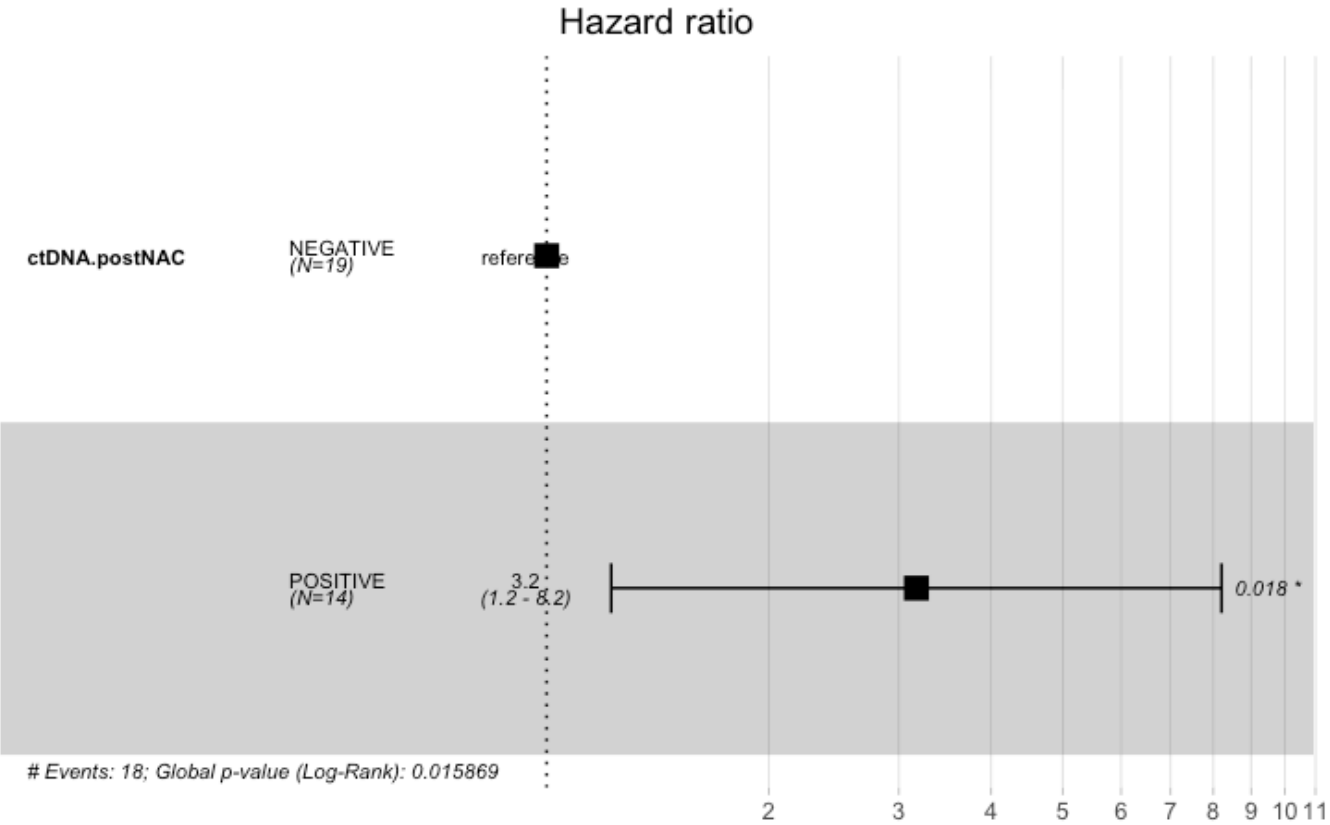
```
summary(KM_curve, times= c(24))
```

Call: `survfit(formula = surv_object ~ ctDNA.postNAC, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

ctDNA.postNAC=NEGATIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
04	24.000	8.000	7.000	0.632	0.111	0.379	0.8
ctDNA.postNAC=POSITIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
79	24.0000	2.0000	11.0000	0.2143	0.1097	0.0521	0.44

Hide

```
circ_data$ctDNA.postNAC <- factor(circ_data$ctDNA.postNAC, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.postNAC, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:
coxph(formula = surv_object ~ ctDNA.postNAC, data = circ_data)

n= 33, number of events= 18

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.postNACPOSITIVE	1.1532	3.1684	0.4857	2.375	0.0176 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.postNACPOSITIVE	3.168	0.3156	1.223	8.208

Concordance= 0.65 (se = 0.055)
Likelihood ratio test= 5.82 on 1 df, p=0.02
Wald test = 5.64 on 1 df, p=0.02
Score (logrank) test = 6.27 on 1 df, p=0.01

Hide


```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 3.17 (1.22-8.21); p = 0.018"
```

#EFS by ctDNA Clearance post-NAT

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.C2D1.Clearance <- NA #first we create the variable for the ctDNA & NAC combination, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.C2D1.Clearance = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" ~ "TRUE",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" ~ "FALSE",
  ))

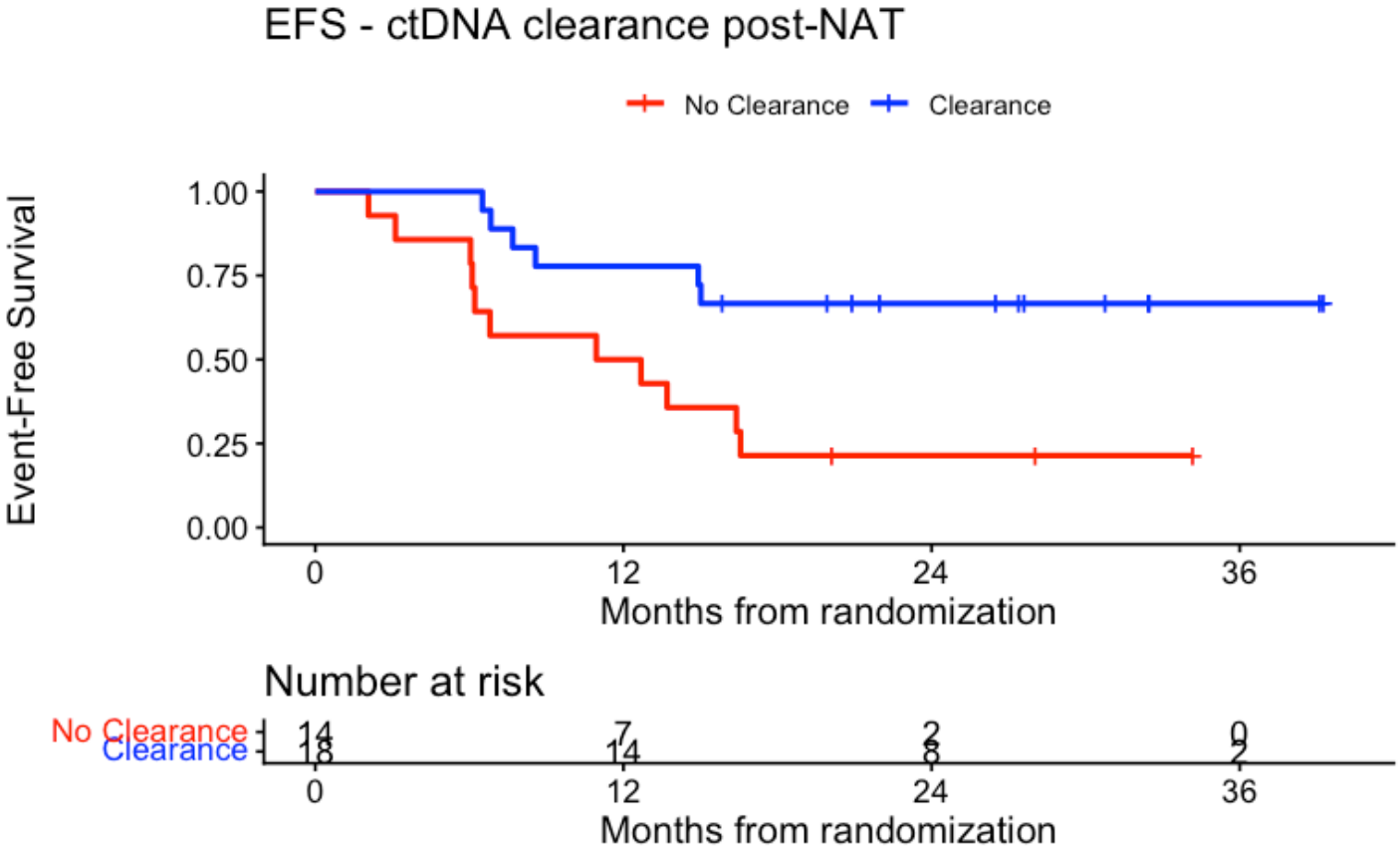
circ_data <- circ_data[!is.na(circ_data$ctDNA.C2D1.Clearance),]
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.C2D1.Clearance, data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$EFS.months, event = circ_data\$EFS.Event) ~ ctDNA.C2D1.Clearance, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.C2D1.Clearance=FALSE	14	11	11.8	6.21	NA
ctDNA.C2D1.Clearance=TRUE	18	6	NA	15.01	NA

Hide

```
surv_object <- Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.C2D1.Clearance, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=12, palette=c("red","blue"), title="EFS - ctDNA clearance post-NAT", ylab="Event-Free Survival", xlab="Months from randomization", legend.labs=c("No Clearance", "Clearance"), legend.title="")
```



Hide

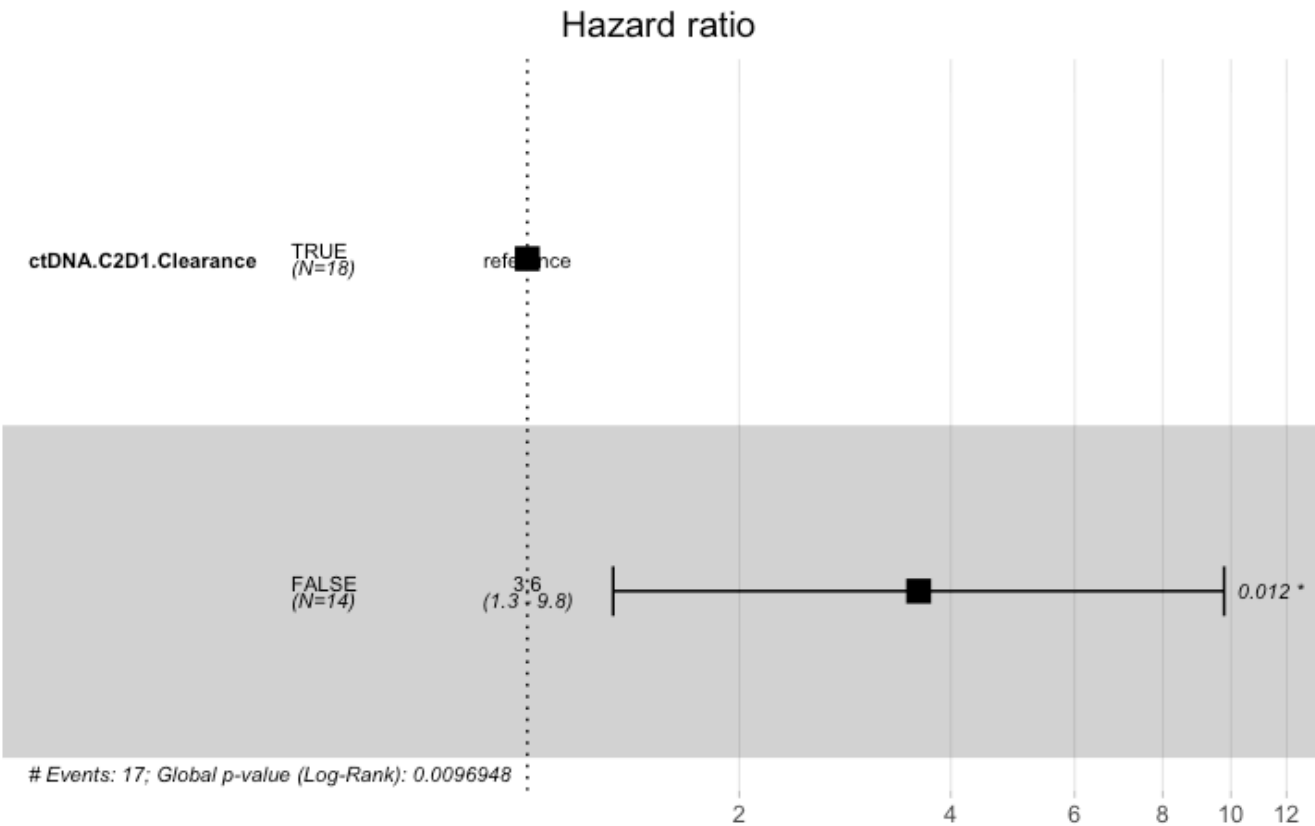
```
summary(KM_curve, times= c(24))
```

```
Call: survfit(formula = surv_object ~ ctDNA.C2D1.Clearance, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

ctDNA.C2D1.Clearance=FALSE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
79	24.0000	2.0000	11.0000	0.2143	0.1097	0.0521	0.44
ctDNA.C2D1.Clearance=TRUE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
34	24.000	8.000	6.000	0.667	0.111	0.404	0.8

Hide

```
circ_data$ctDNA.C2D1.Clearance <- factor(circ_data$ctDNA.C2D1.Clearance, levels=c("TRUE", "FALSE"))
cox_fit <- coxph(surv_object ~ ctDNA.C2D1.Clearance, data=circ_data)
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.C2D1.Clearance, data = circ_data)
```

n= 32, number of events= 17

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.C2D1.ClearanceFALSE	1.2809	3.5997	0.5102	2.51	0.0121 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.C2D1.ClearanceFALSE	3.6	0.2778	1.324	9.785

Concordance= 0.665 (se = 0.055)

Likelihood ratio test= 6.69 on 1 df, p=0.01

Wald test = 6.3 on 1 df, p=0.01

Score (logrank) test = 7.18 on 1 df, p=0.007

Hide

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 3.6 (1.32-9.78); p = 0.012"
```

#Association of ctDNA status with Pathological N status

Hide

```
rm(list=ls())
```

```
setwd("~/Downloads")
```

```
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
```

```
circ_data$ctDNA.postNAC <- factor(circ_data$ctDNA.postNAC, levels = c("NEGATIVE", "POSITIVE"), labels = c("Negative", "Positive"))
```

```
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c("N0", "N1-N3"))
```

```
contingency_table <- table(circ_data$pN.stage, circ_data$ctDNA.postNAC)
```

```
chi_square_test <- chisq.test(contingency_table)
```

```
print(chi_square_test)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: contingency_table
X-squared = 3.4977, df = 1, p-value = 0.06145
```

[Hide](#)

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.04638
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.9734627 471.1789080
sample estimates:
odds ratio
 9.250939
```

[Hide](#)

```
print(contingency_table)
```

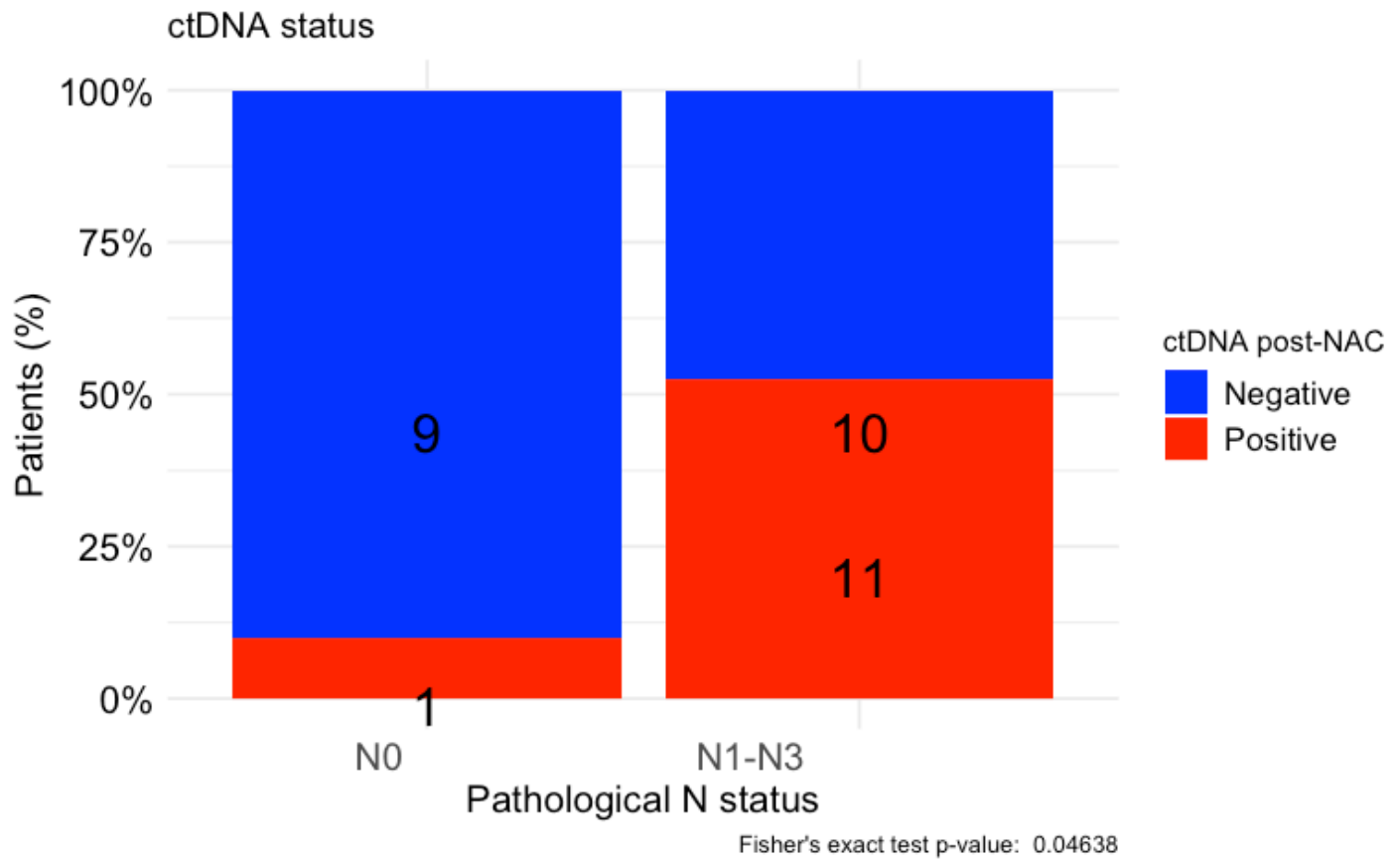
	Negative	Positive
N0	9	1
N1-N3	10	11

[Hide](#)

```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA status",
        x = "Pathological N status",
        y = "Patients (%)",
        fill = "ctDNA post-NAC",
        caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Negative" = "blue", "Positive" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
        legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



#Association of ctDNA clearance with Pathological N status

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.clearance <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.clearance = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" ~ "TRUE",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" ~ "FALSE",
  ))

circ_data <- circ_data[!is.na(circ_data$ctDNA.clearance),]

circ_data$ctDNA.clearance <- factor(circ_data$ctDNA.clearance, levels = c("TRUE", "FALSE"), labels = c("Clearance", "No Clearance"))
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c("N0", "N1-N3"))
contingency_table <- table(circ_data$pN.stage, circ_data$ctDNA.clearance)
chi_square_test <- chisq.test(contingency_table)
print(chi_square_test)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: contingency_table
X-squared = 2.9167, df = 1, p-value = 0.08767
```

[Hide](#)

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.04921
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.8452809 424.1258917
sample estimates:
odds ratio
 8.235925
```

[Hide](#)

```
print(contingency_table)
```

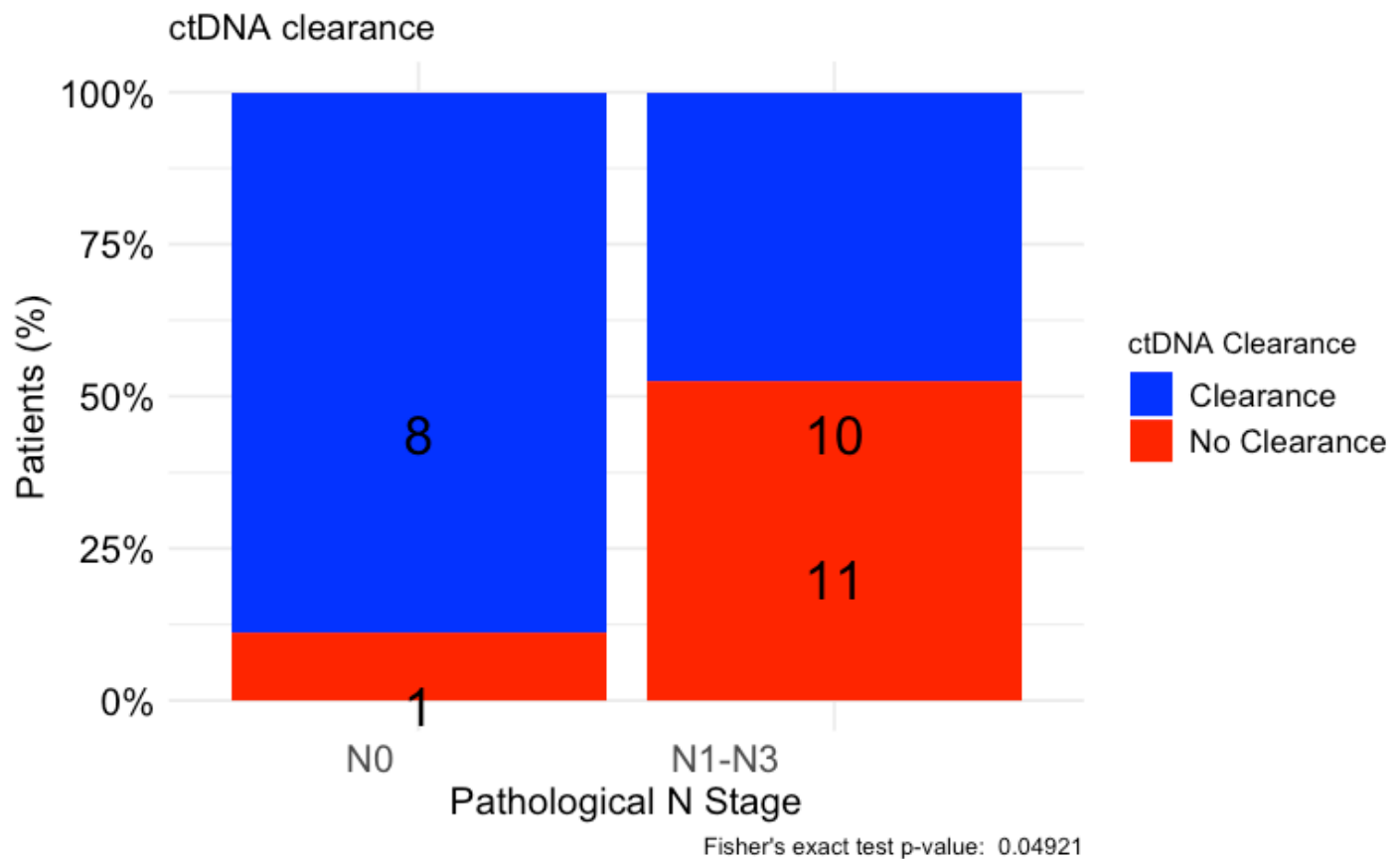
	Clearance	No Clearance
N0	8	1
N1-N3	10	11

[Hide](#)


```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA clearance",
        x = "Pathological N Stage",
        y = "Patients (%)",
        fill = "ctDNA Clearance",
        caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Clearance" = "blue", "No Clearance" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
        legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



#Association of ctDNA dynamics with Pathological N status

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.Dynamics <- NA #first we create the variable for the ctDNA & NAC combina
tion, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" & ctDNA.MRD == "NEGATIVE" ~
    "ctDNA +/-/-",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" & ctDNA.MRD == "NEGATIVE" ~
    "ctDNA +/+/-",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" & ctDNA.MRD == "POSITIVE" ~
    "ctDNA +/+/"
  ))

circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("ctDNA +/-/-", "ct
DNA +/+/-", "ctDNA +/+/"))
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c
("N0", "N1-N3"))
contingency_table <- table(circ_data$ctDNA.Dynamics, circ_data$pN.stage)
chi_square_test <- chisq.test(contingency_table)
print(chi_square_test)
```

Pearson's Chi-squared test

```
data: contingency_table
X-squared = 7.8431, df = 2, p-value = 0.01981
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.01785
alternative hypothesis: two.sided
```

Hide

```
print(contingency_table)
```

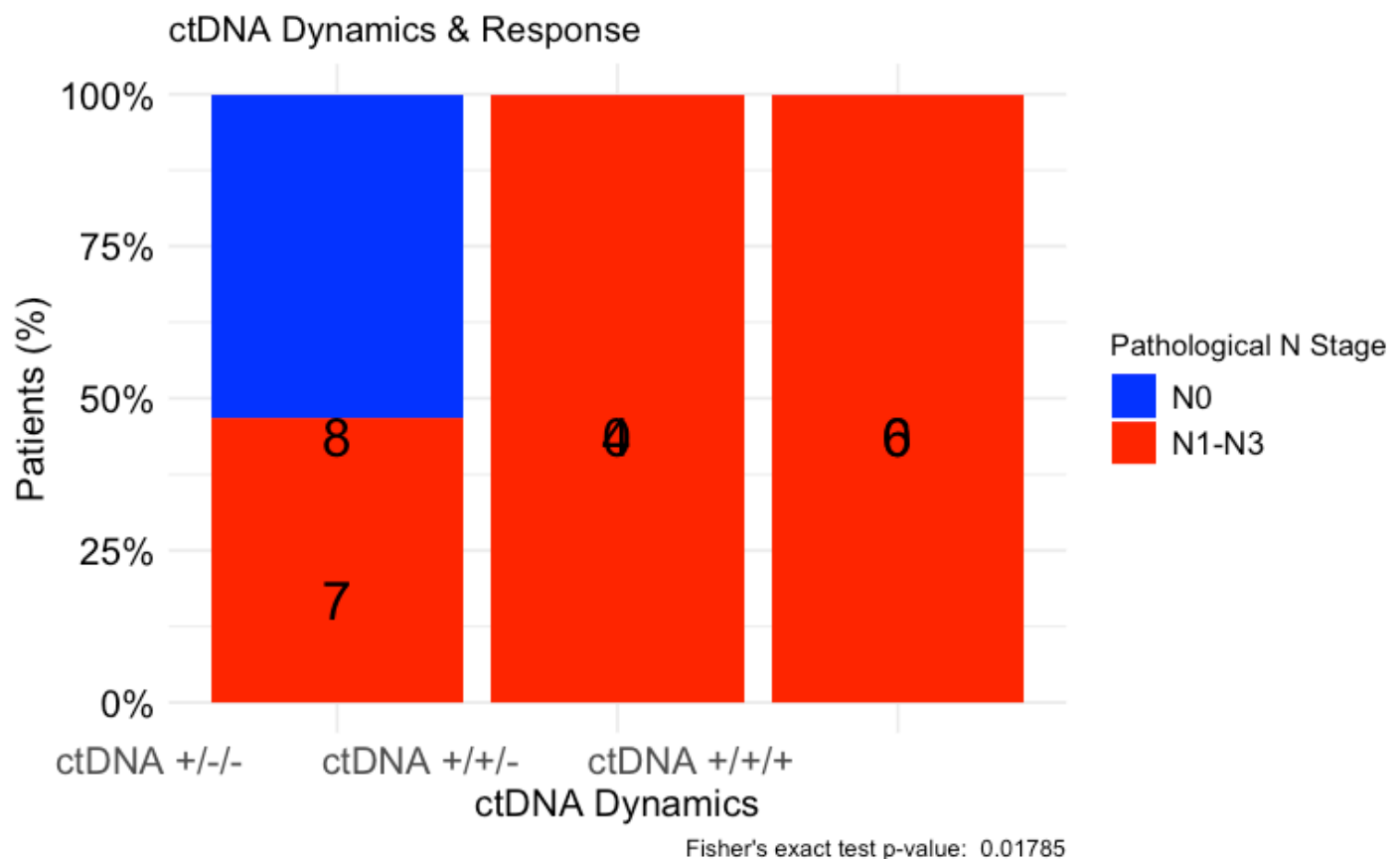
	N0	N1-N3
ctDNA +/-/-	8	7
ctDNA +/+/-	0	4
ctDNA +/+/+	0	6

Hide

```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black",
    vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA Dynamics & Response",
    x = "ctDNA Dynamics",
    y = "Patients (%)",
    fill = "Pathological N Stage",
    caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("N0" = "blue", "N1-N3" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
    text size
    axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
    axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
    axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
    legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



#EFS by ctDNA at the MRD Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.MRD!="",]

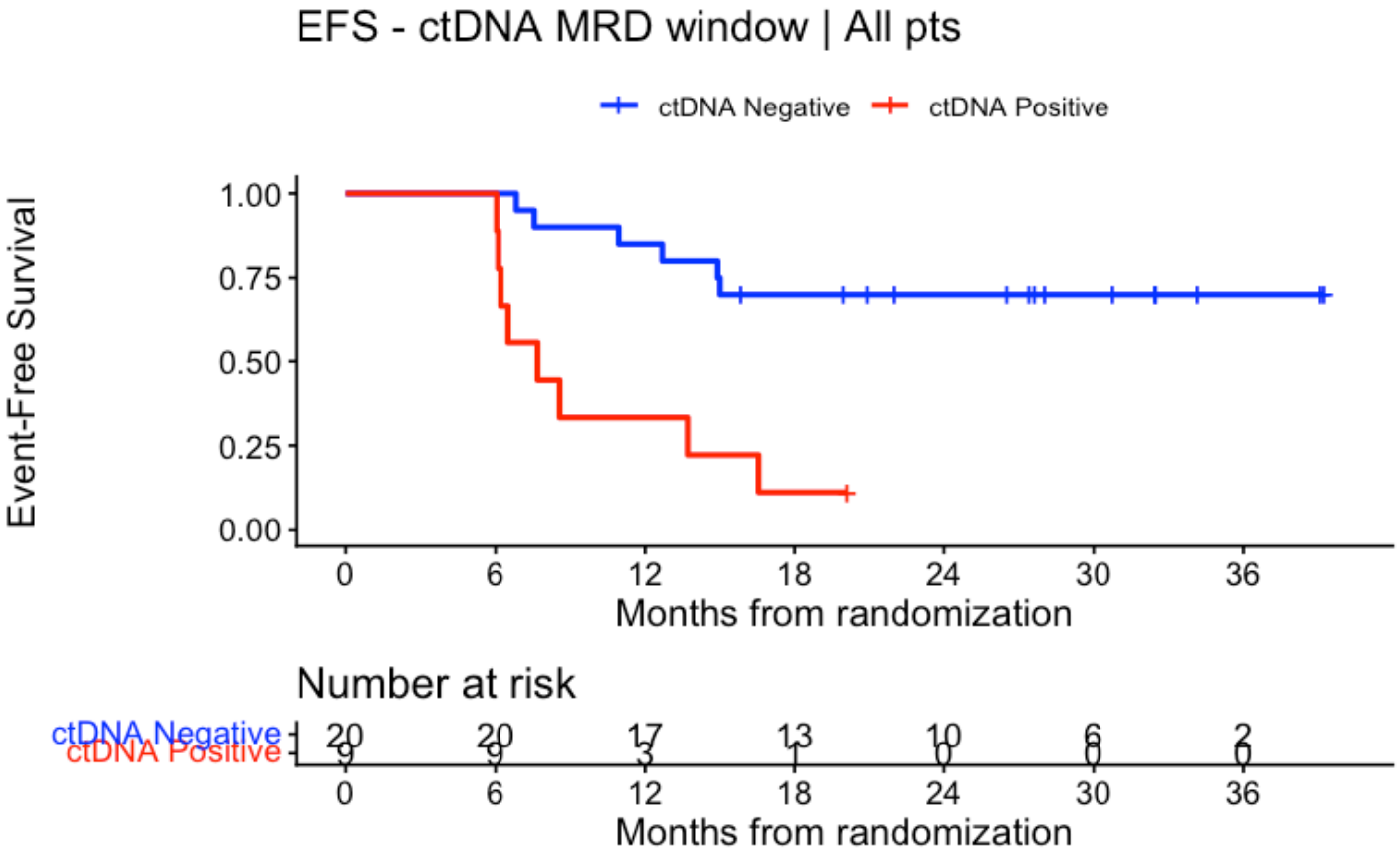
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.MRD, data =
circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$EFS.months, event = circ_data\$EFS.Event) ~
ctDNA.MRD, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.MRD=NEGATIVE	20	6	NA	NA	NA
ctDNA.MRD=POSITIVE	9	8	7.69	6.21	NA

Hide

```
surv_object <-Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.MRD, data = circ_data,conf.int=0.95,conf.type="l
og-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA MRD window | All pts", y
lab= "Event-Free Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negat
ive", "ctDNA Positive"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(0, 24))
```

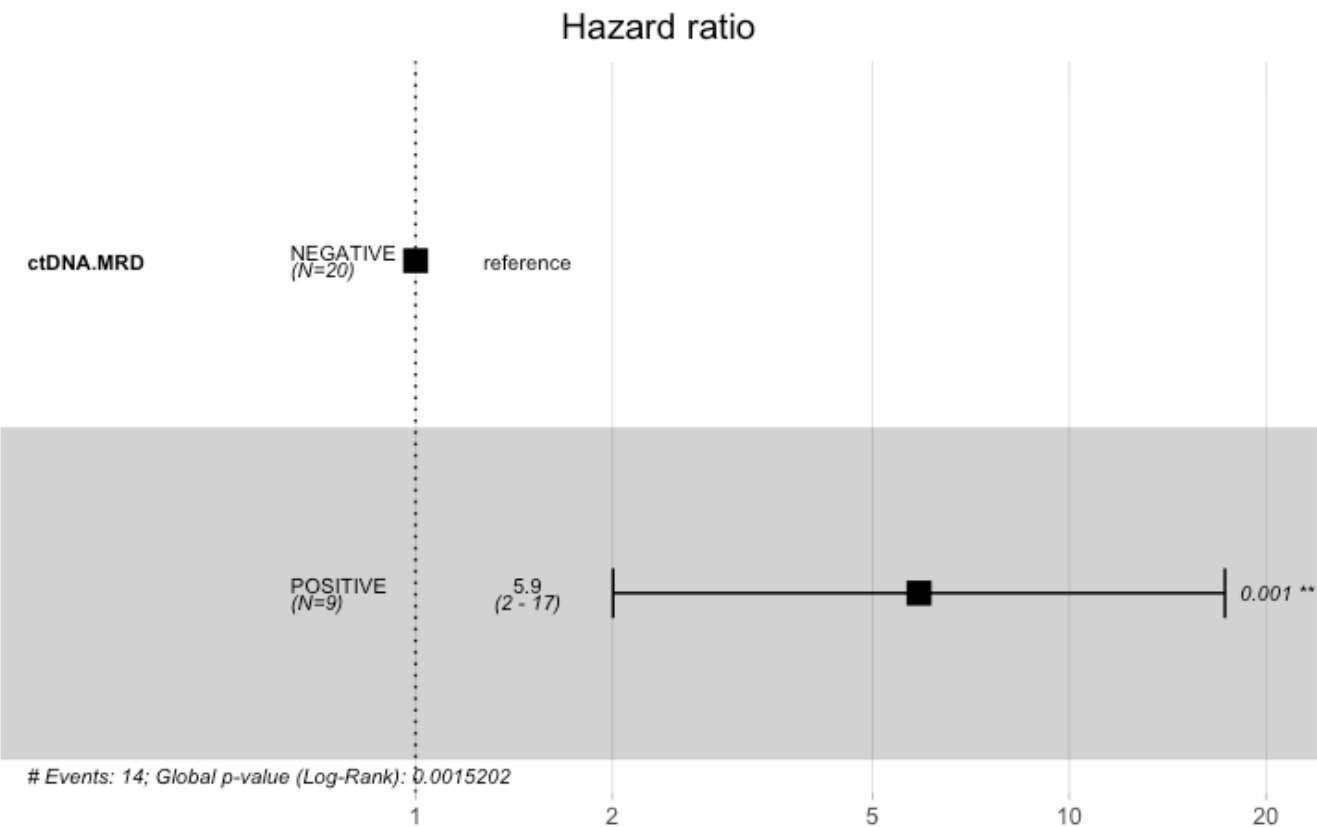
```
Call: survfit(formula = surv_object ~ ctDNA.MRD, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

ctDNA.MRD=NEGATIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
0	20	0	1.0	0.000	1.000	1.000	
24	10	6	0.7	0.102	0.451	0.853	

ctDNA.MRD=POSITIVE							
CI	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	0	9	0	1	0	1	

Hide

```
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.MRD, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.MRD, data = circ_data)
```

n= 29, number of events= 14

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.MRDPOSITIVE	1.7732	5.8898	0.5498	3.225	0.00126 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.MRDPOSITIVE	5.89	0.1698	2.005	17.3

Concordance= 0.708 (se = 0.055)

Likelihood ratio test= 10.05 on 1 df, p=0.002

Wald test = 10.4 on 1 df, p=0.001

Score (logrank) test = 13.14 on 1 df, p=3e-04

Hide

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 5.89 (2.01-17.3); p = 0.001"
```

#OS by ctDNA at the MRD Window

Hide

```
rm(list=ls())
```

```
setwd("~/Downloads")
```

```
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
```

```
circ_data <- circ_data[circ_data$ctDNA.MRD!="",]
```

```
survfit(Surv(time = circ_data$OS.months, event = circ_data$OS.Event)~ctDNA.MRD, data = circ_data)
```



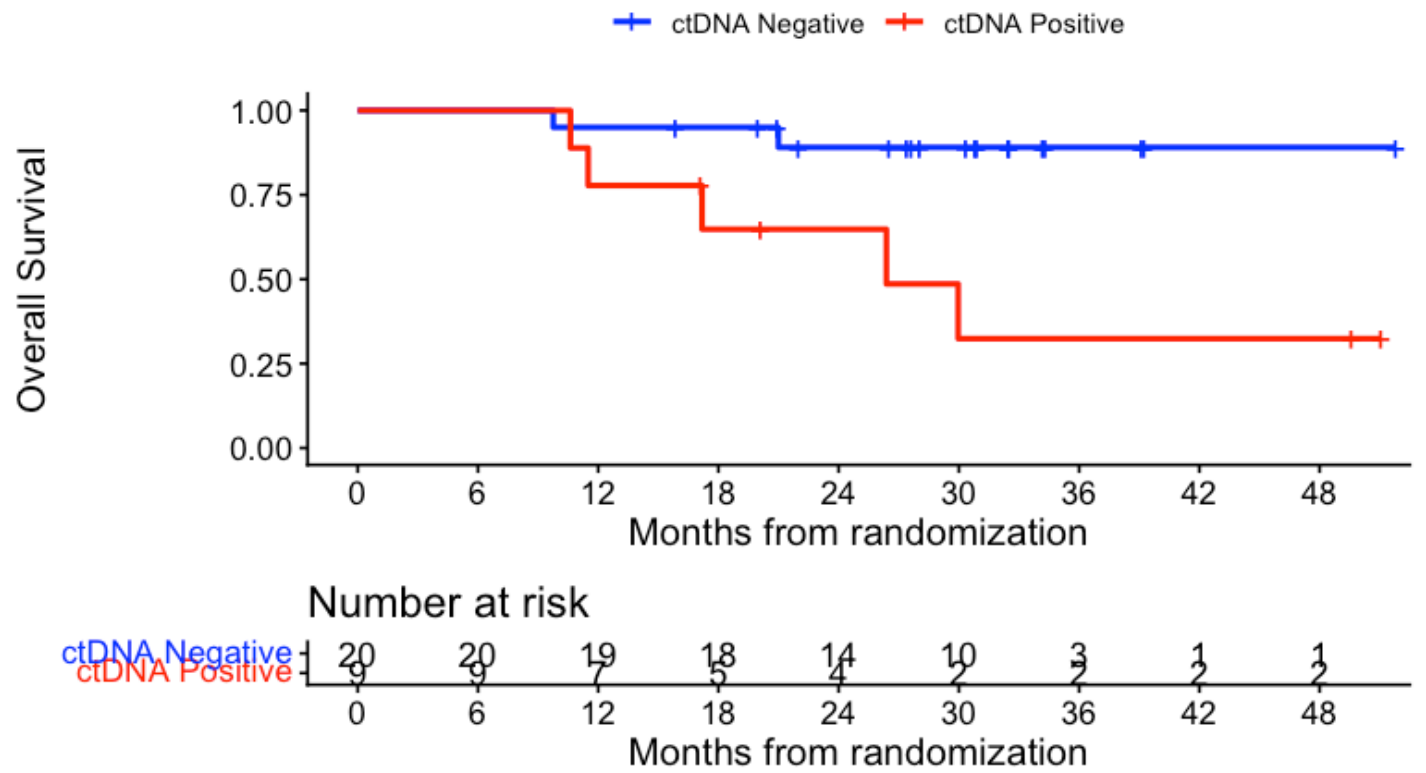
```
Call: survfit(formula = Surv(time = circ_data$OS.months, event = circ_data$OS.Event) ~
      ctDNA.MRD, data = circ_data)
```

	n	events	median	0.95LCL	0.95UCL
ctDNA.MRD=NEGATIVE	20	2	NA	NA	NA
ctDNA.MRD=POSITIVE	9	5	26.4	17.2	NA

[Hide](#)

```
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.MRD, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="OS - ctDNA MRD window | All pts", ylab= "Overall Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```

OS - ctDNA MRD window | All pts



Hide

```
summary(KM_curve, times= c(0, 24))
```

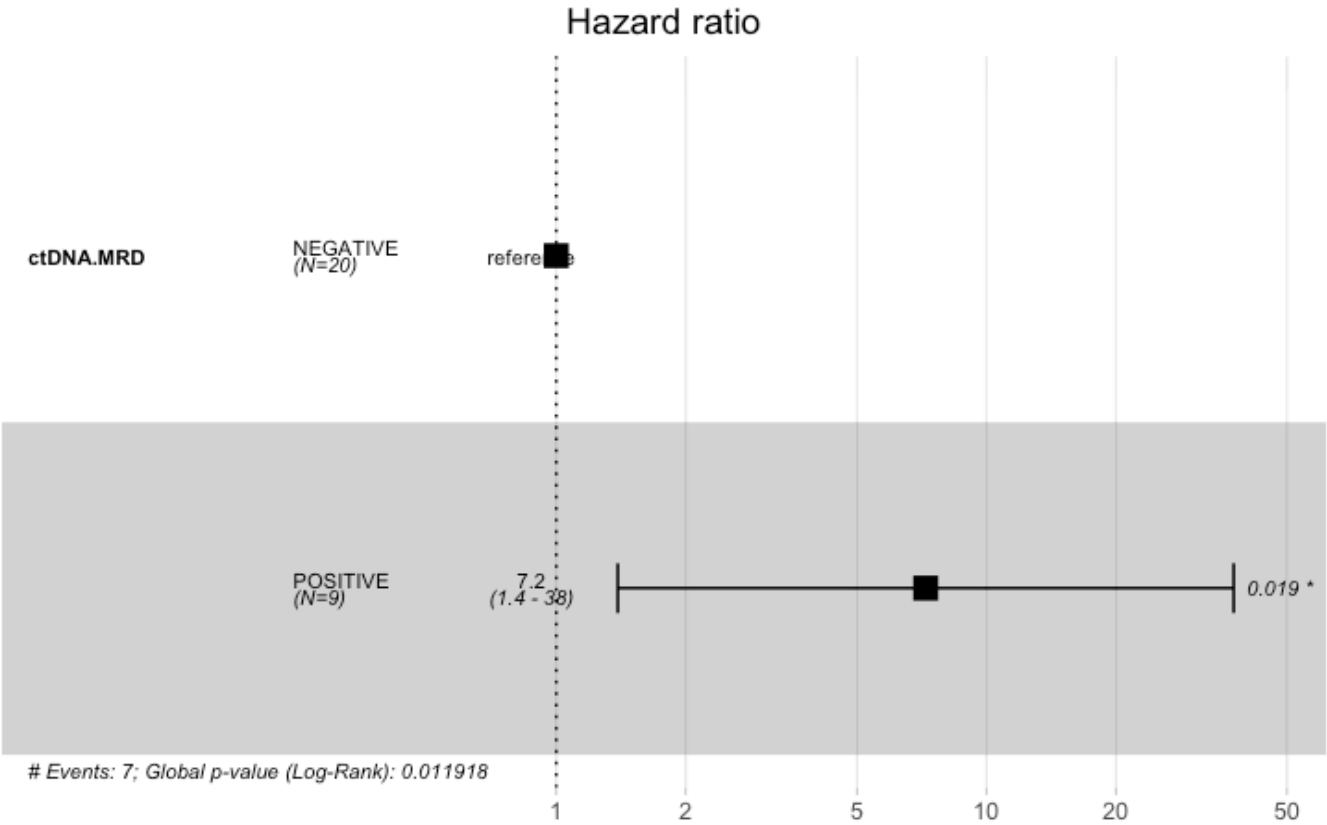
```
Call: survfit(formula = surv_object ~ ctDNA.MRD, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

ctDNA.MRD=NEGATIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
0	20	0	1.000	0.0000	1.000	1.000	
24	14	2	0.891	0.0734	0.627	0.972	

ctDNA.MRD=POSITIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
0	9	0	1.000	0.000	1.000	1.000	
24	4	3	0.648	0.165	0.253	0.872	

Hide

```
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.MRD, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:
coxph(formula = surv_object ~ ctDNA.MRD, data = circ_data)

n= 29, number of events= 7

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.MRDPOSITIVE	1.9779	7.2277	0.8411	2.351	0.0187 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.MRDPOSITIVE	7.228	0.1384	1.39	37.58

Concordance= 0.72 (se = 0.093)
Likelihood ratio test= 6.32 on 1 df, p=0.01
Wald test = 5.53 on 1 df, p=0.02
Score (logrank) test = 7.5 on 1 df, p=0.006

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 7.23 (1.39-37.58); p = 0.019"
```

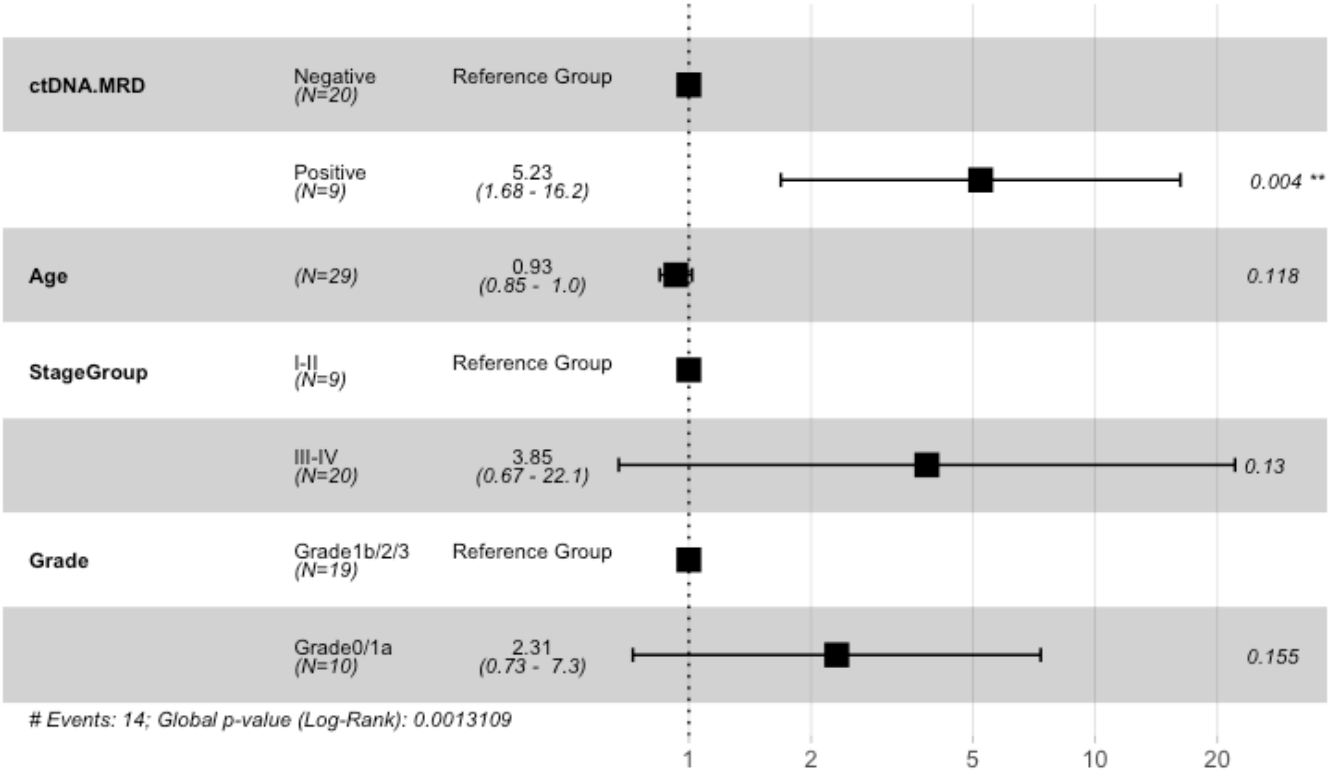
#Multivariate cox regression for PFS - ctDNA at MRD

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.MRD!="",]

circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"), labels = c("Negative", "Positive"))
circ_data$Sex <- factor(circ_data$Sex, levels=c("Female","Male"))
circ_data$ECOG <- factor(circ_data$ECOG, levels = c("0", "1"))
circ_data$Location <- factor(circ_data$Location, levels=c("Ut","Mt","Lt"))
circ_data$StageGroup <- factor(circ_data$StageGroup, levels=c("I-II","III-IV"))
circ_data$Grade <- factor(circ_data$Grade, levels=c("Grade1b/2/3","Grade0/1a"))
surv_object <- Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
cox_fit <- coxph(surv_object ~ ctDNA.MRD + Age + StageGroup + Grade, data=circ_data)
ggforest(cox_fit, data = circ_data, main = "Multivariate Regression Model for PFS", refLabel = "Reference Group")
```

Multivariate Regression Model for PFS



Hide

```
test.ph <- cox.zph(cox_fit)
```